

AI BASED SMART FARM ASSISTANT AND MARKETPLACE

AGRISENSE

Code Ninjas

AGRISENSE

MEET OUR TEAM



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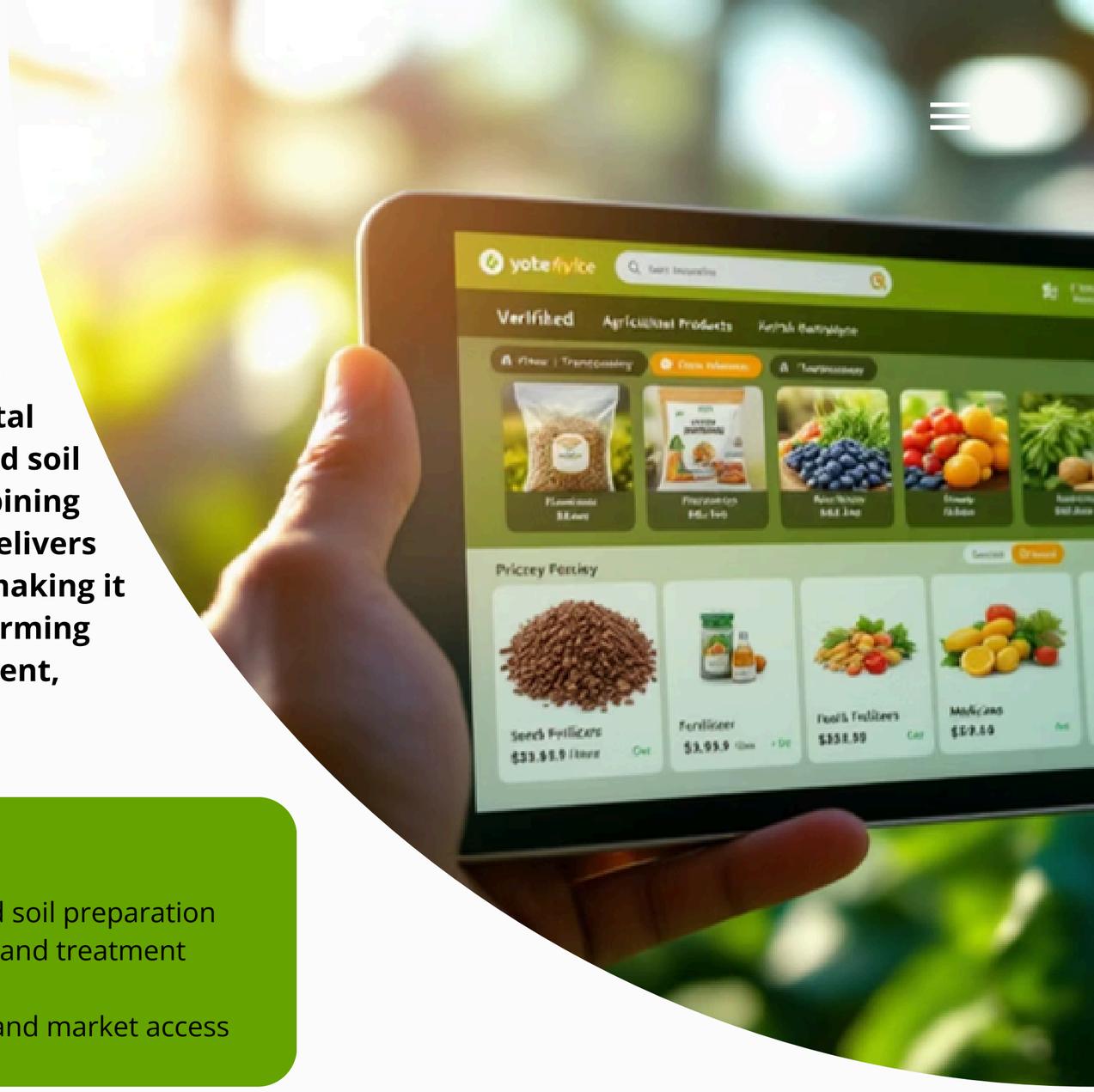


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What is AgriSense?



AgriSense is an AI-based smart Farm Assistant & marketplace platform designed for Bangladeshi farmers. It works as a digital "Krishi Advisor" that supports farmers from crop selection and soil preparation to disease detection and market access. By combining artificial intelligence with local agricultural data, AgriSense delivers accurate guidance in Bangla using voice and visual support, making it easy to use even for low-literacy users. Its goal is to reduce farming uncertainty, prevent crop loss, and help farmers make confident, data-driven decisions.

Accessibility First

- they can't read text and don't get timely advice on crop disease and weather.
- digital krishi advisor to help farmer using simple voice and visual guide in bangla

Holistic Support

- Crop selection and soil preparation
- Disease detection and treatment advice
- Harvest planning and market access

Proposed - Requirements

01

Voice-First AI Assistant (Smart Farming Voice Agent)

- Allows hands-free operation using simple Bangla voice commands
- Converts spoken queries to action via Speech-to-Text & LLM
- Understand their voice
- gives instant audio feedback in natural Bangla language

02

Snap & Heal (AI Disease Detective)

- Farmer takes a photo of a sick plant
- AI detects the disease
- Provides treatment suggestions in Bangla
- Recommends locally available medicines

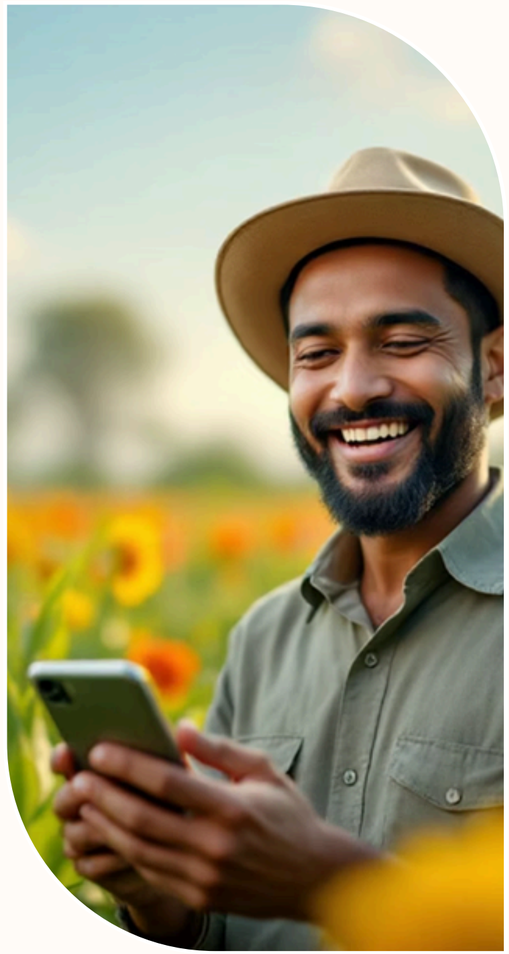
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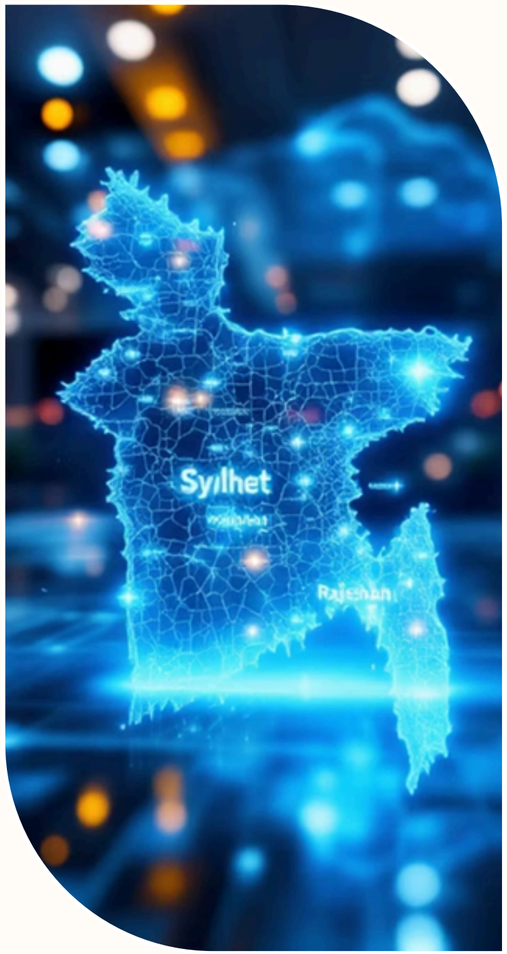
Soil-Soul Match (Smart Crop Selector)

- Analyzes regional soil types
  - Barind clay soil
  - Sandy soil in coastal Chars
- Recommends the most suitable crops
- Optimized for Kharif and Rabi seasons



Proposed - Requirements



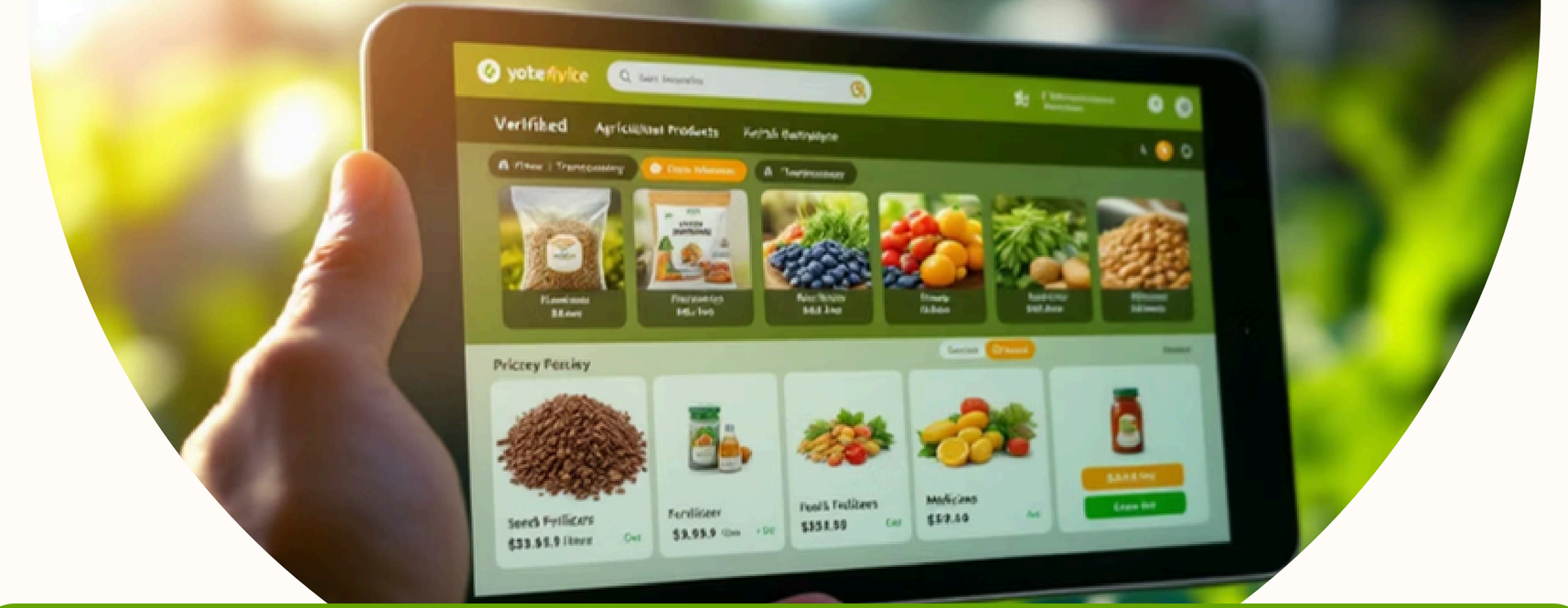


04 Shovon-Boli (Voice Assistant)

- Short answer in Bangla text and voice, taken from verified documents.
- Typing queries, or traveling to find an extension officer for simple questions.

05 Treatment to product matching

- A ranked list of matching products from verified vendors.
- Searching for the right product and seller by oneself.



Haat_Bazar: The Smart Marketplace

The Bridge to Action

A specialised marketplace offering access to verified seeds, high-quality fertilisers, and essential agricultural medicines.

Fair Trade Assurance

Protects farmers from middleman exploitation through real-time price transparency and accurate market rate tracking.

Transparency & Trust

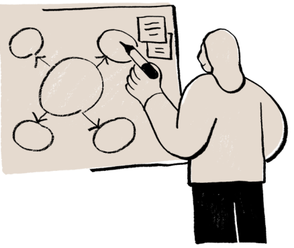
Features visual order tracking (Pending → Shipped → Delivered), providing clear updates and building confidence.

This innovative platform ensures farmers receive fair prices for their produce and have reliable access to vital resources.

Field Research and Stakeholder Consultation



Meet Limon Bhai, A proud son of a farming family, deeply rooted in agriculture.







Nazmul Bhai, A young agricultural entrepreneur, turning farming into an opportunity for growth.



Prokash Bhai, A local shopkeeper and part-time farmer with firsthand experience in agriculture.



Field Research and Stakeholder Consultation



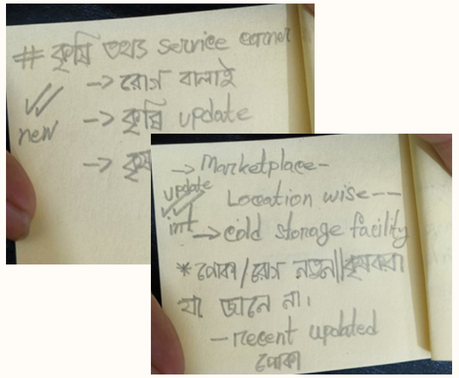
Meet Kbd Ahsan Habib Bhai — An agricultural consultant and researcher dedicated to advancing agricultural knowledge, with a B.Sc. in Agriculture and an M.Sc. in Entomology from Gazipur Agricultural University. His expertise brings valuable scientific insights to our mission of building smarter, more sustainable farming solutions.



Gives us Suggestions

Valuable scientific insights

New Features



REQUIREMENT CHANGES AFTER ELICITATION



The system shall detect diseases from leaf photos.

The system shall enable online order payments.

Voice control covered all key platform features.

MODIFIED

The system shall detect leaf diseases, show confidence scores, and request recapture or expert review below 0.70.

Orders support cash on delivery or direct online payments between buyers and sellers.

Krishi Barta is added to the screens that the farmer can open, listen to and control by voice.

REQUIREMENT CHANGES AFTER ELICITATION

Demands

Farmers need updated, expert-verified Bangla agricultural news, district alerts, voice reading, and offline access.

Added Requirments

The system shall provide Krishi Barta, an auto-updated news section in Bangla where farmers receive expertverified information on new diseases, insect pests, viruses and nutrient-deficiency disorders, with district-based alerts, voice reading and offline access.

Demands

Farmers need voice-controlled navigation and confirmed voice-based product ordering, including recommended pesticides.

Added Requirments

The system shall let the farmer control the app by voice from any screen: speech is converted to text, the intent is recognized and the requested feature is opened. Let the farmer order a product by voice, including the pesticide recommended after a diagnosis, and shall place the order only after confirmation.



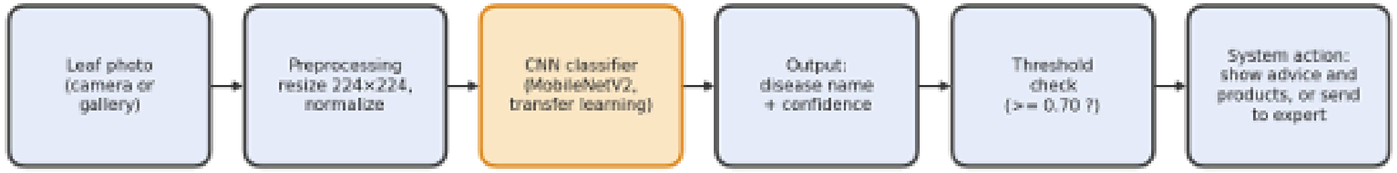
REQUIREMENT CHANGES AFTER ELICITATION

Demands	Added Requirments
Farmers need Bangla voice queries answered using verified knowledge.	**The system shall allow farmers to ask agricultural questions in Bangla using voice and receive accurate, reliable answers from a verified knowledge base, ensuring accessible and trustworthy farming guidance.**
Demands	**Added Requirments**
Photos captured offline are queued and uploaded when internet returns.	**The system shall store photos captured offline in a queue and automatically uploads them to the server when internet access returns, allowing farmers to continue capturing crop images without interruption and receive disease diagnosis once connected.**

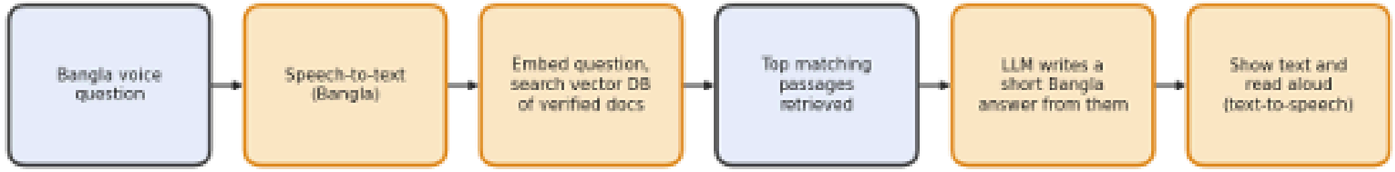


AI Feature Integration

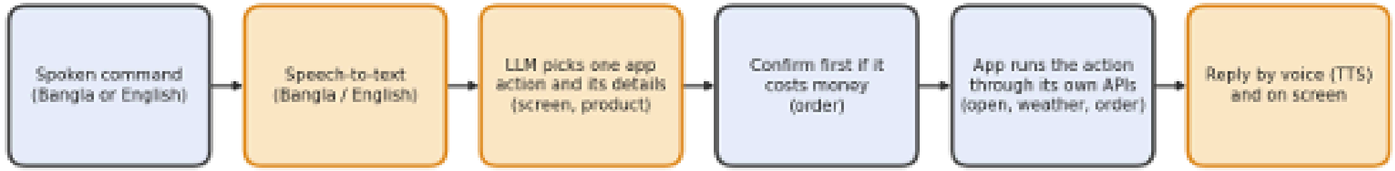
A. Disease detection pipeline



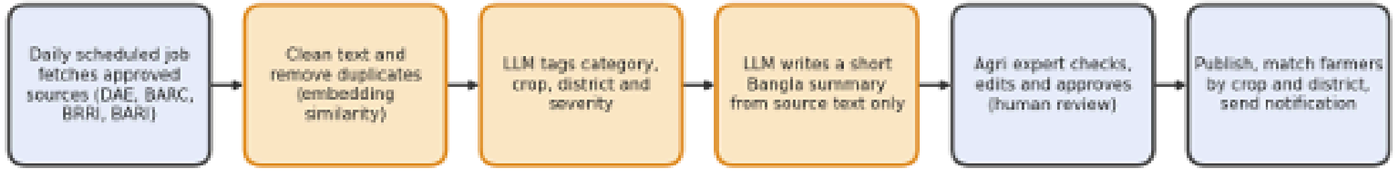
B. Voice assistant pipeline (RAG)



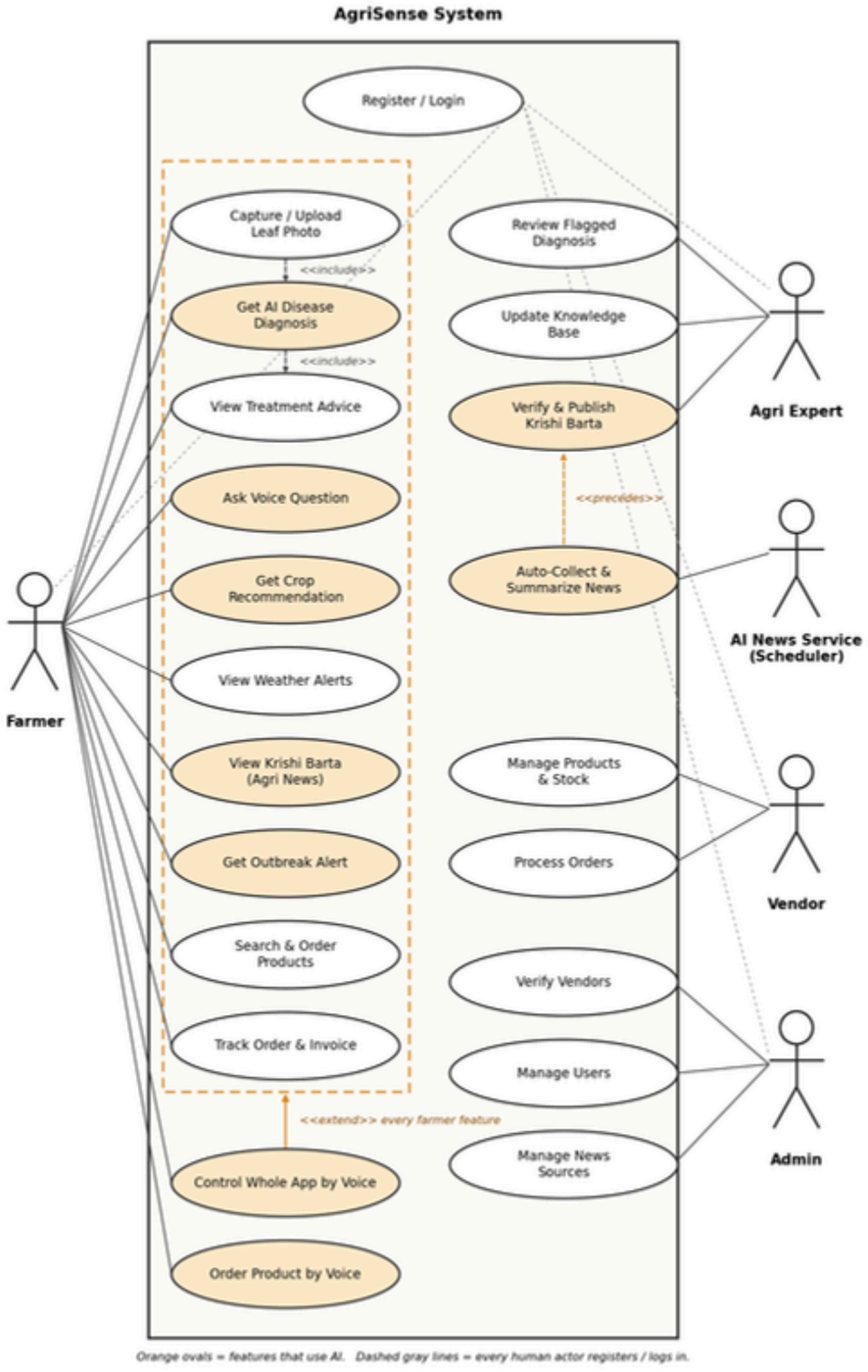
C. Voice command pipeline (app control)



D. Krishi Barta news pipeline (scheduled)



Orange boxes = AI steps.



User Roles & System Interaction

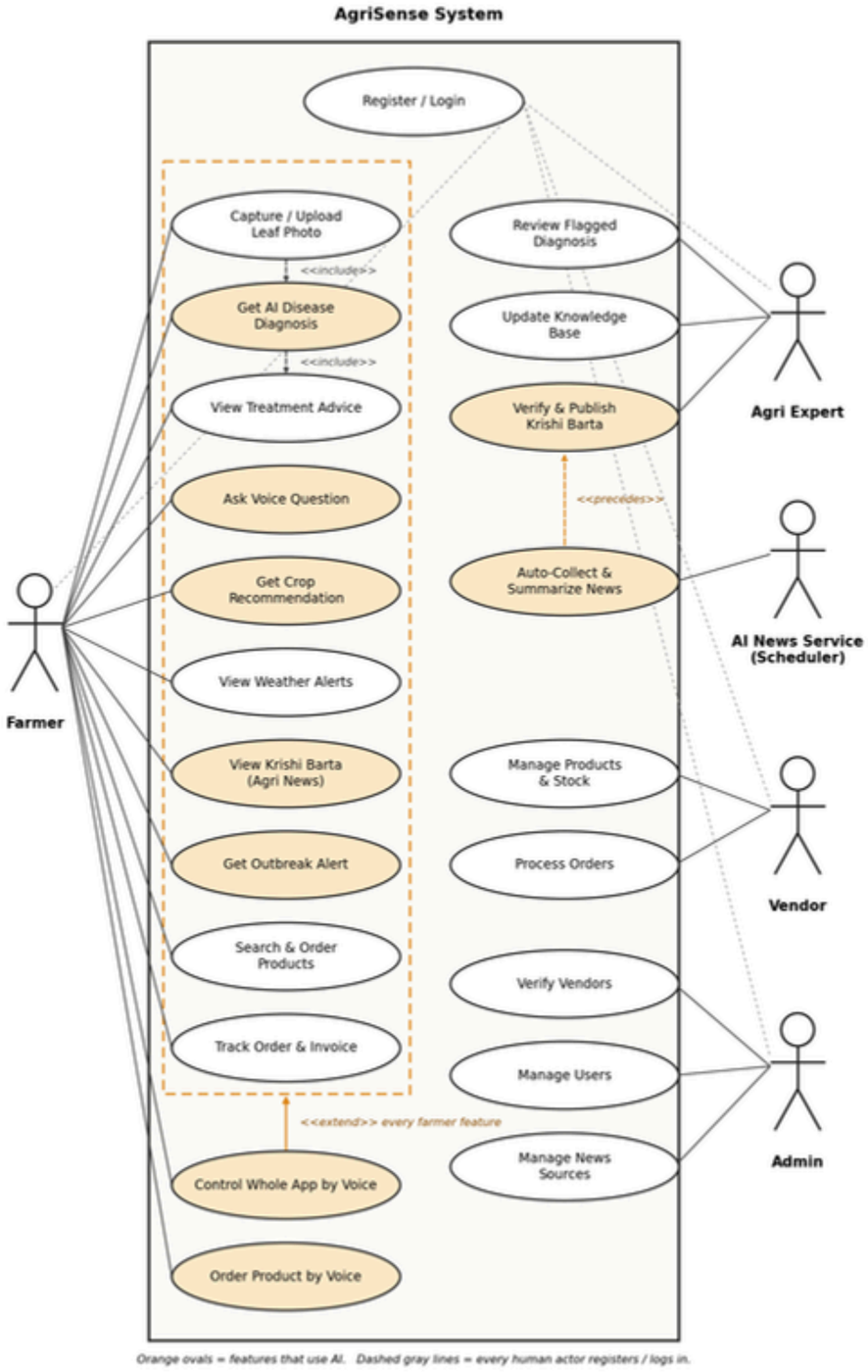
Farmer: Uploads leaf photos, receives AI diagnosis, uses voice search, and orders products.

Vendor: Manages product inventory and processes customer orders.

Agri Expert: Reviews flagged AI diagnoses and updates the expert knowledge base.

Admin: Handles vendor verification and manages overall user security.

Access: All users access the system via a unified Register / Login process



Core AI & Diagnostic Loop

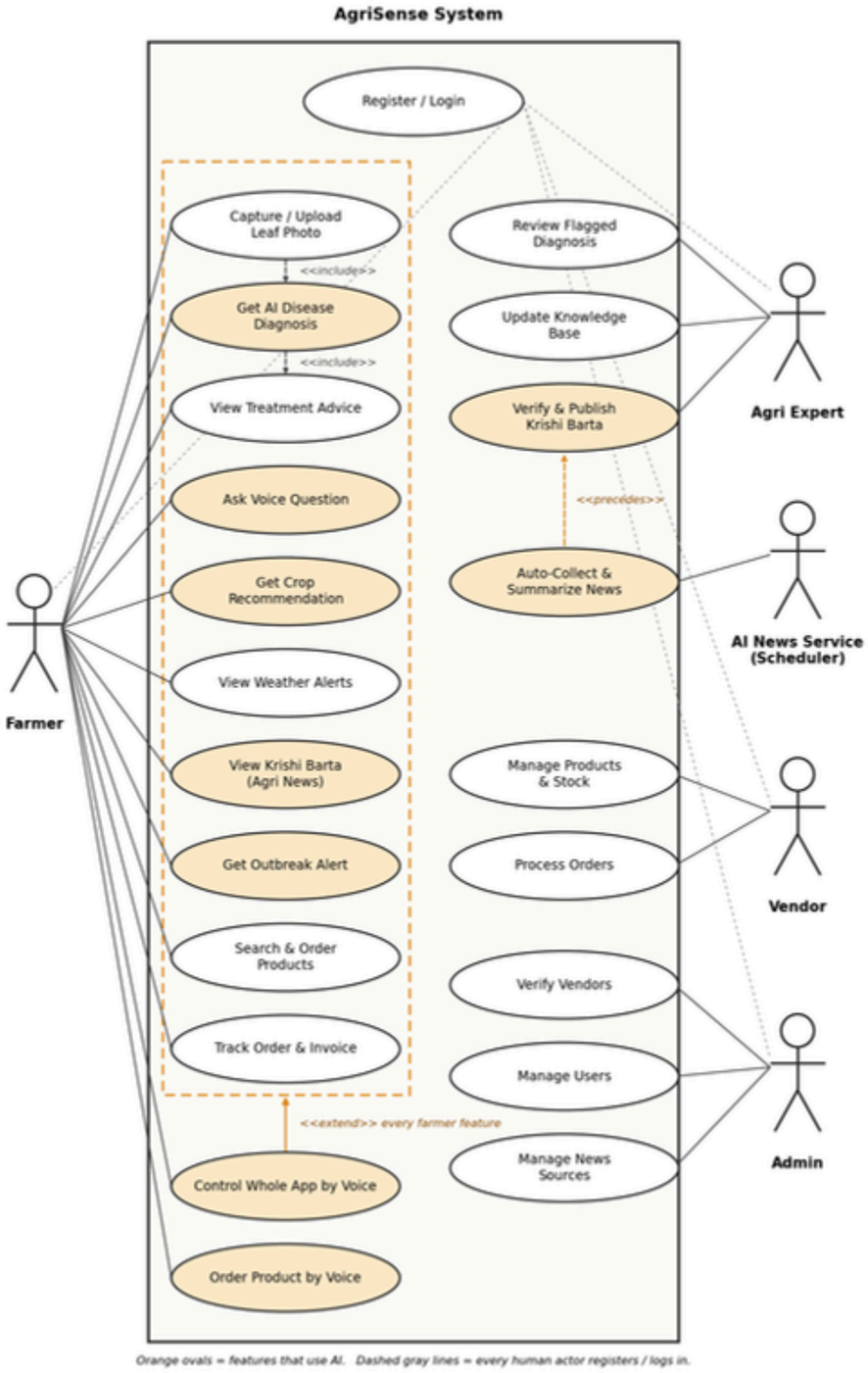
Diagnostic Loop: Capture Photo → AI Diagnosis → Treatment Advice.

AI Features (Orange Ovals):
AI Disease Diagnosis: Automated plant illness detection.

Ask Voice Question: Hands-free voice assistant for farmers.

Get Crop Recommendation: Smart advice to improve yield.





Marketplace & Accessibility

E-Commerce Flow: Farmers search products → Vendors process orders → Digital invoice & tracking generated.

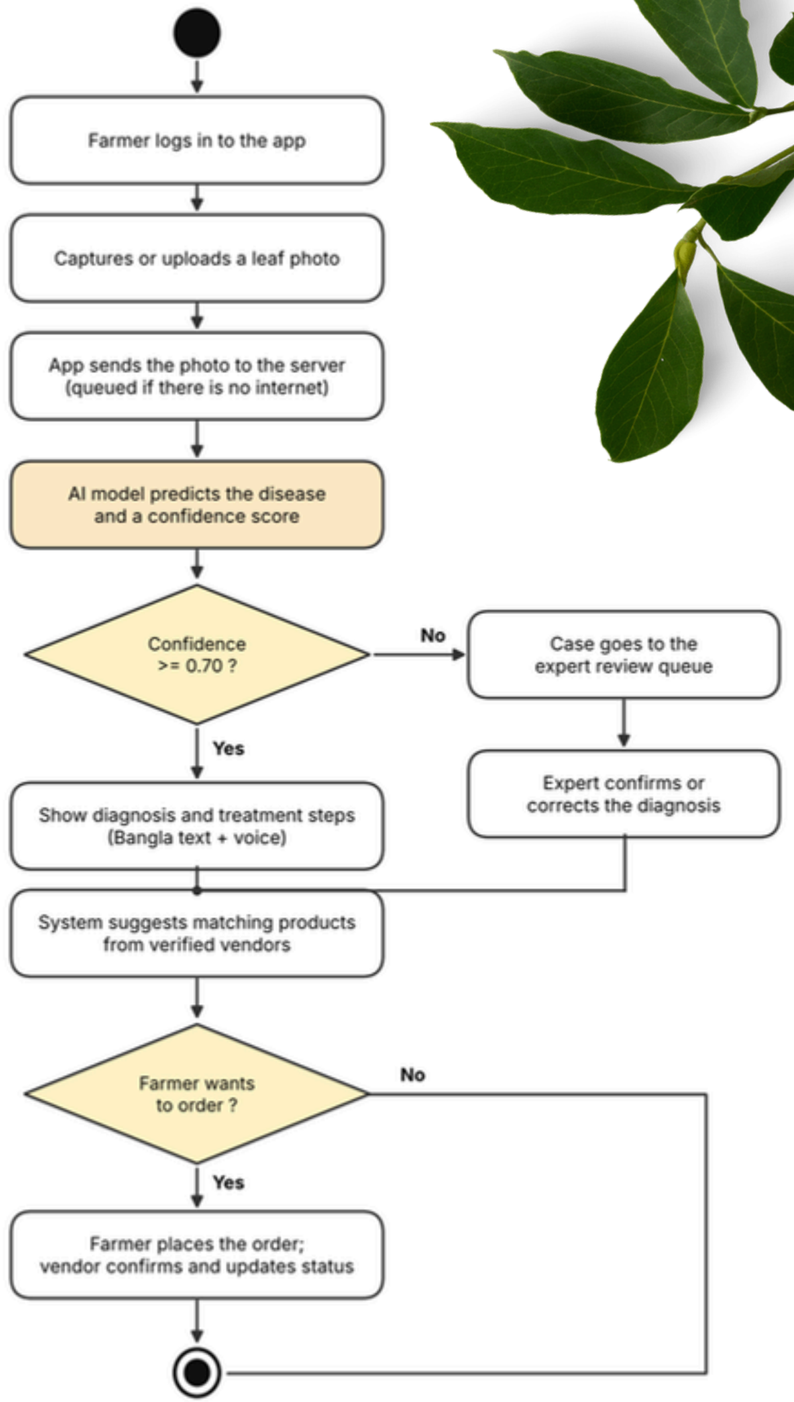
Low Literacy Support: Built-in AI voice assistance enables easy navigation for all farmers.



Activity Diagram

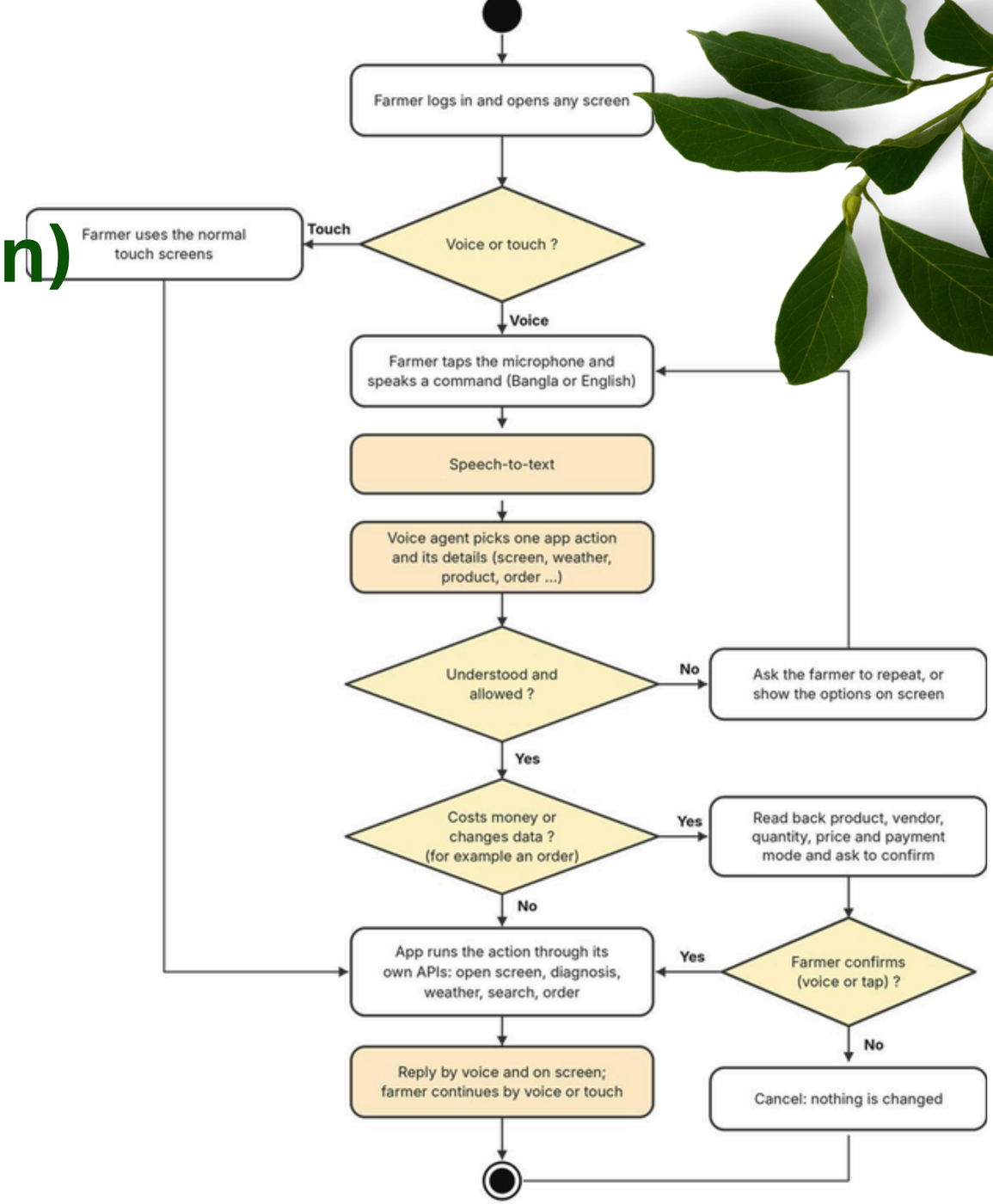
1. AI-Powered Advisory:
   - Farmer uploads leaf photo (queued offline if needed).
   - AI model provides rapid diagnosis with confidence score and Bangla text/voice treatment steps.
   - Low confidence cases (< 0.70) are automatically routed for expert review.

2. Unified E-Commerce Loop:
   - System suggests validated treatments and products from verified vendors.
   - Farmer can immediately place a product order based on the diagnosis.
   - Vendors confirm the order and update fulfillment status.



Activity Diagram (Voice-Controlled Interaction)

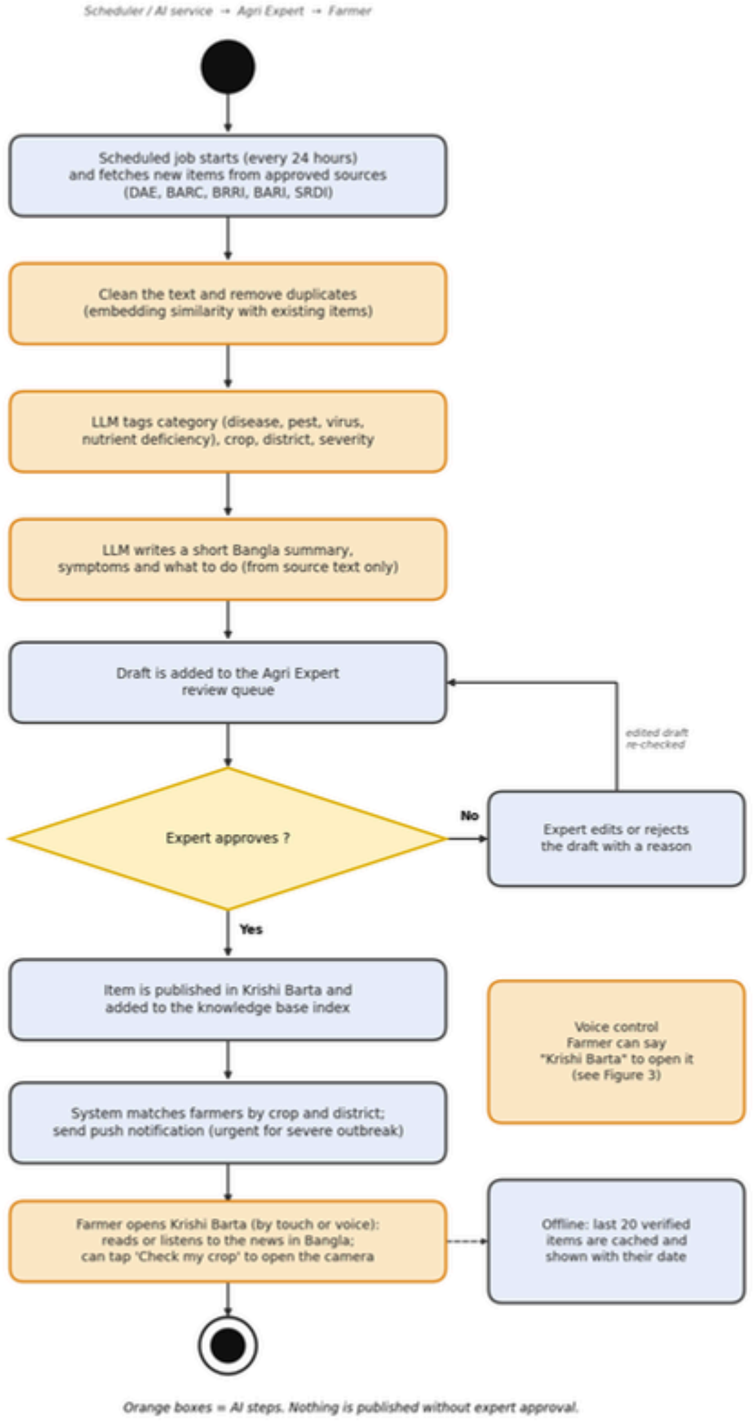
- Voice or touch: The farmer speaks in Bangla or English, and the touch path always stays available.
- AI understanding: Speech-to-text converts the command, and the voice agent picks one app action. If unclear, the app asks the farmer to repeat or shows options.
- Safe confirmation: Orders and data changes are read back (product, vendor, quantity, price, payment) and need a voice or tap confirmation.
- Reliable execution: The app runs actions through its own APIs and replies by voice and on screen.
-



Orange boxes = AI steps. The touch path always stays available.

Activity Diagram(Krishi Barta)

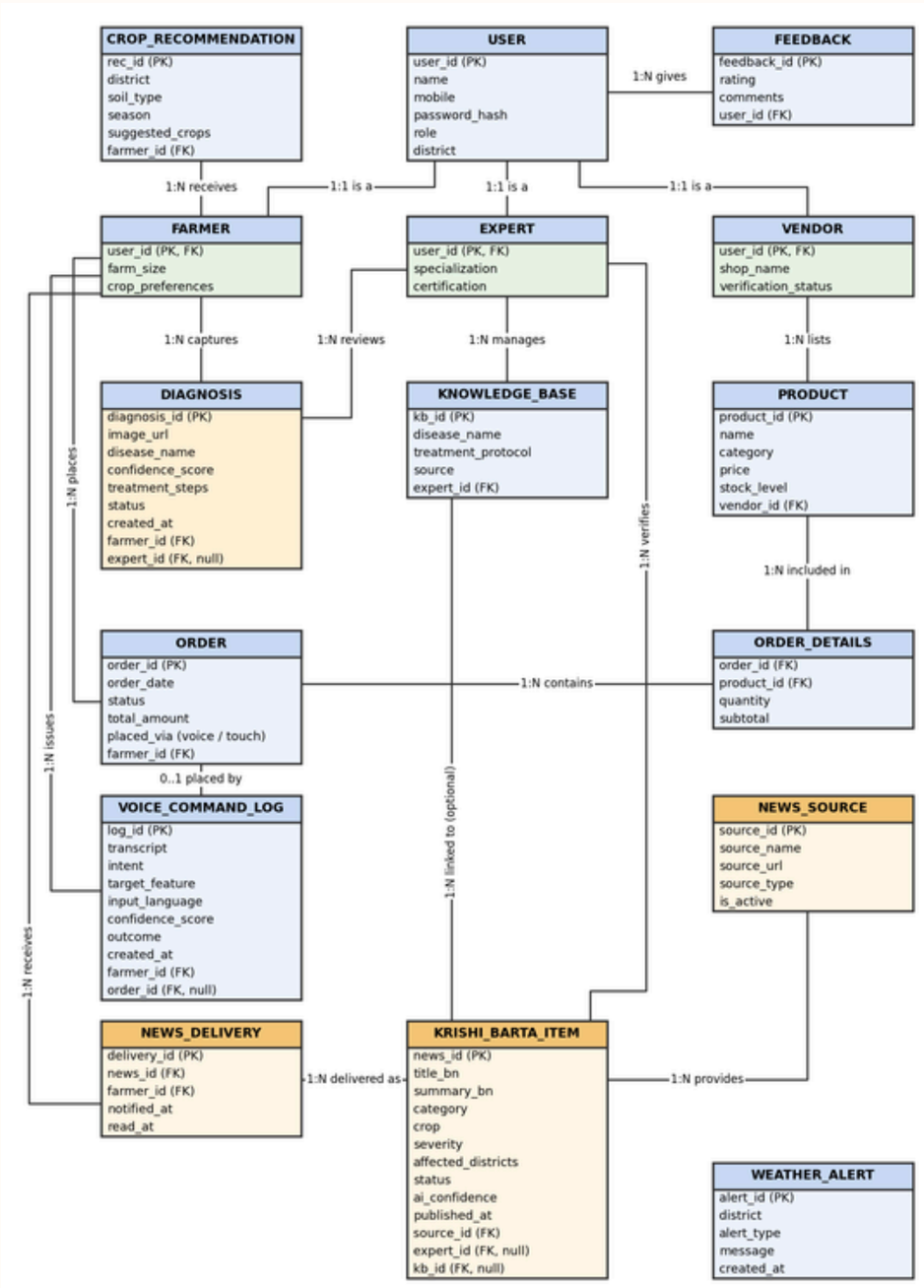
- Automated collection: A daily job fetches new items from approved sources (DAE, BARC, BRRI, BARI, SRDI) and removes duplicates.
- AI processing: The LLM tags category, crop, district and severity, then writes a short Bangla summary from the source text only.
- Expert-in-the-loop: Every draft is reviewed by an Agri Expert. Nothing is published without approval.
- Targeted delivery: Farmers are matched by crop and district and get push notifications, urgent for severe outbreaks.
- Farmer experience: Read or listen in Bangla by touch or voice, tap “Check my crop” to open the camera, and use the last 20 items offline.
-



ENTITY RELATIONSHIP DIAGRAM

Data Foundation

- **Unified user model:** One USER table with 1:1 role specialization into FARMER, EXPERT and VENDOR.
- **Core advisory data:** DIAGNOSIS stores image, confidence score and treatment steps. Low-confidence cases link to an expert for review.
- **Commerce data:** ORDER and PRODUCT form an M:N relationship, resolved through ORDER_DETAILS, with orders linked to verified vendors.
- **New for Krishi Barta:** KRISHI_BARTA_ITEM, NEWS_SOURCE and NEWS_DELIVERY (orange tables) support verified agri-news delivery to farmers.
- **Voice support:** VOICE_COMMAND_LOG records transcript, intent, language and outcome. Voice-placed orders link back to ORDER.




AGRISENSE
THANK YOU